OSPF asoslari

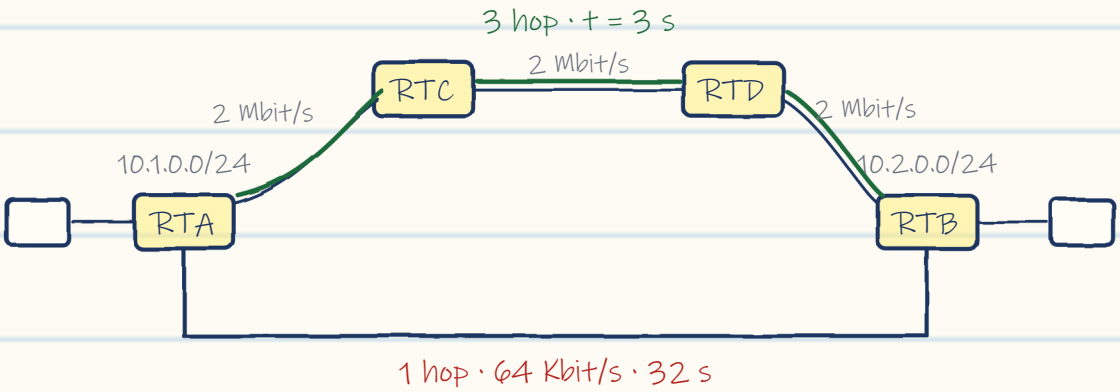
OSPF Basics

Asosiy atamalar

- OSPF — link-state ichki shlyuz protokoli (IGP), RFC 2328
- LSA — tarmoq holati haqidagi e'lon — qo'shnilarga tarqaladi
- LSDB — LSA lardan yig'ilgan tarmoq xaritasi bazasi
- SPF — Dijkstra algoritmi — xaritadan eng qisqa yo'l
- Cost — interfeys metrikasi; yo'l narxi — costlar yig'indisi

1. RIP nega yetmaydi

- hop sanaydi — polosa/kechikish hisobga olinmaydi
- 16 hop = cheksiz — tarmoq diametri kichik
- sekin konvergensiya — suppress timer ushlab turadi
- 30 s da butun jadval: 1000 marshrut ≈ 160 KB

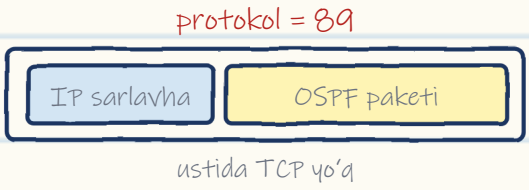


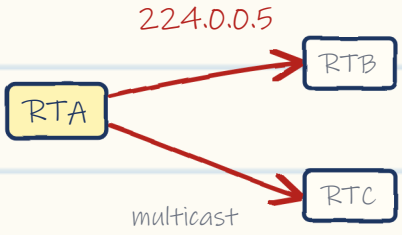
1-rasm. RIP 1 hop tanlaydi: 32 s, uzun yo'l — 3 s

2. OSPF nima

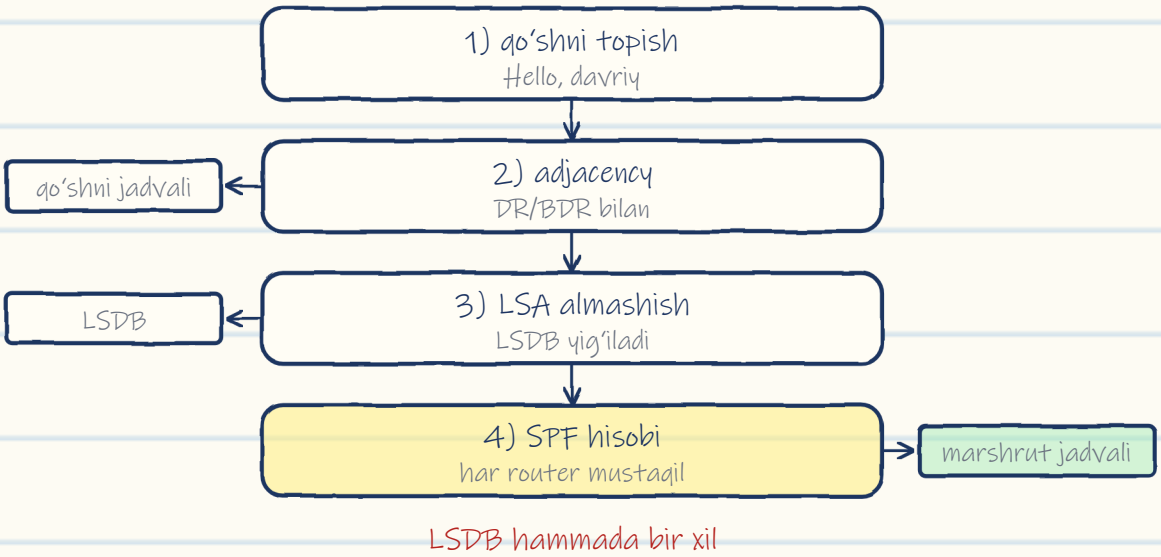
Link-state ichki shlyuz protokoli, IETF RFC 2328 = OSPFv2

- yo'lni SPF (Dijkstra) hisoblaydi
- IP ichida, protokol 89 — ishonchlilikni o'zi ta'minlaydi
- paketlar multicast 224.0.0.5 ga ketadi
- inkremental — faqat so'ralgan LSA



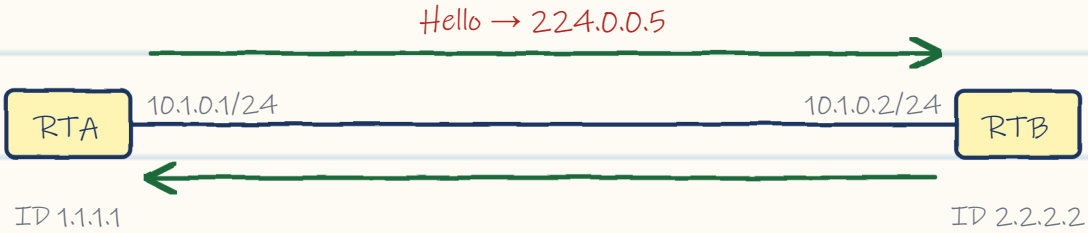


3. To'rt bosqich



- adjacency — nuqta-nuqta virtual kanal
- LSA lar yig'ilib LSDB — butun topologiya

4. Hello paketi



Hello maydonlari

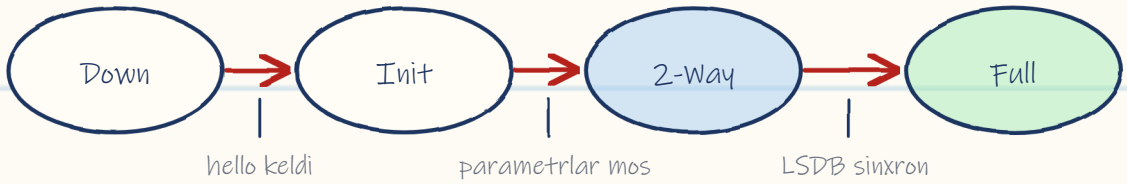
Router ID	Area ID	maska	DR/BDR	prioritet

qo'shni jadvali

Router ID	manzil	holat
2.2.2.2	10.1.0.2	2-Way

Qo'shni manzili = hello'ning manba IP si; routerni AS ichida Router ID tanitadi — loopback qulay.

5. Qo'shni holatlari

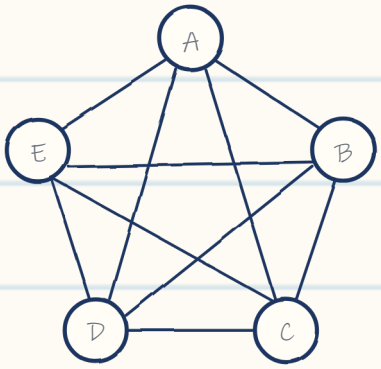


DR/BDR bilan — Full

DRother bilan — 2-Way

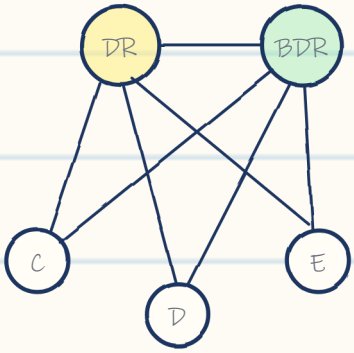
2-rasm. Holat qo'shni jadvalida yoziladi

6. Nega DR/BDR kerak



DR yo'q: n(n-1)/2 = 10

hamma bilan hamma



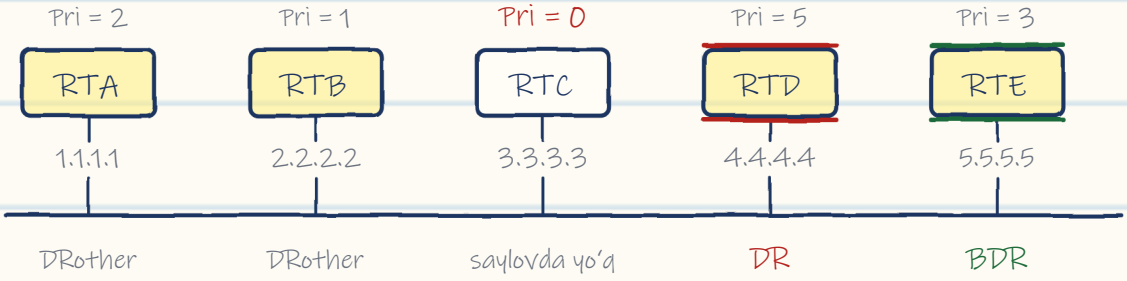
DR/BDR: 2(n-2)+1 = 7

faqat DR/BDR bilan

Routerlar	DR yo'q	DR/BDR bilan
5	10	7
10	45	17

DR interfeysga tegishli, routerga emas: bir tarmoqda DR, boshqasida DRother.

7. DR/BDR saylovi

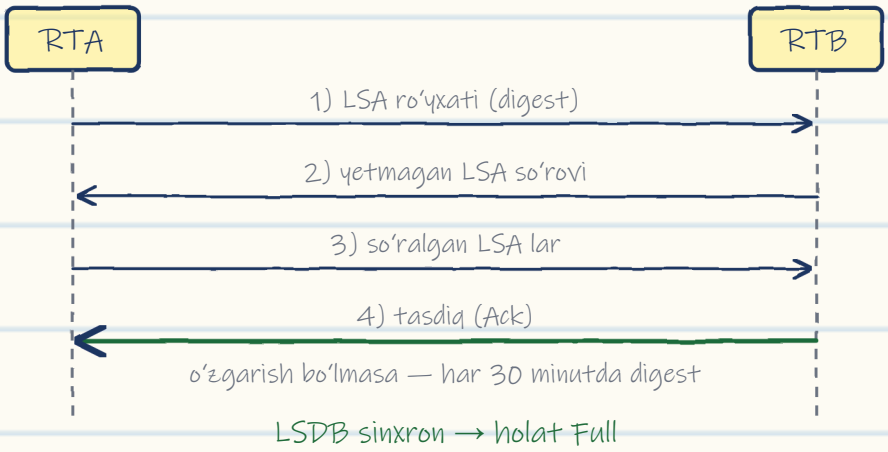


avval BDR, keyin DR saylanadi

keyin kelgan yuqori prioritetli router DR bo'lolmaydi

Qoida: prioritet 0...255; 0 — qatnashmaydi · teng bo'lsa katta Router ID yutadi · da'vogar yo'q — BDR DR bo'ladi.

8. LSA almashinuvi



Ketma-ket raqam — takror paket ishlanmaydi; javob kelmasa qayta yuboriladi. LSA maska tashiydi → VLSM.

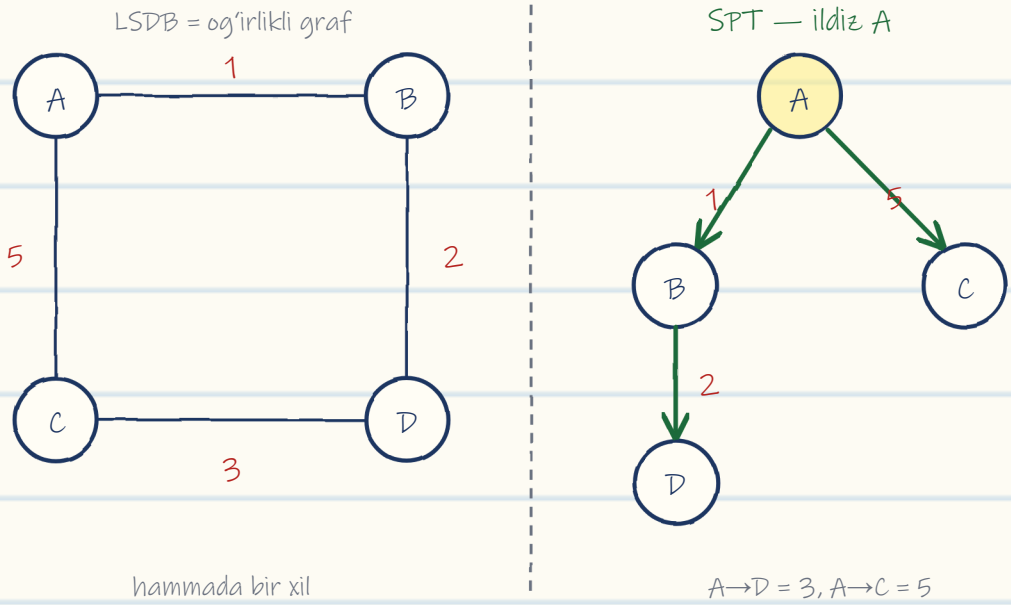
9. Cost — interfeys metrikasi

cost = 100 ÷ polosa (Mbit/s)

polosa	hisob	cost
64 kbit/s	100 ÷ 0,064	1562
2 Mbit/s	100 ÷ 2	50
100 Mbit/s	100 ÷ 100	1

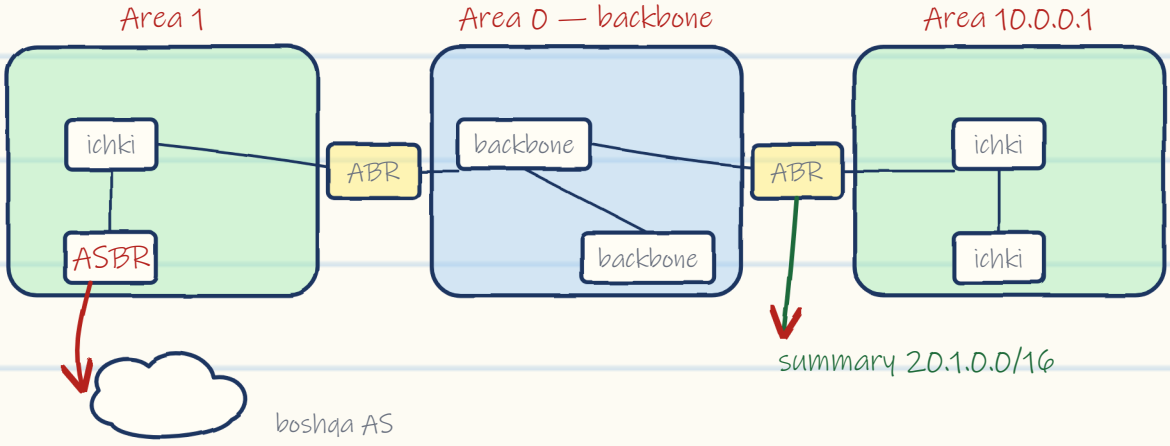
Yo'l narxi = chiquvchi interfeyslar cost yig'indisi; qo'lda ospf cost value.

10. SPF daraxti



- og'irlikli yo'naltirilgan graf, ildiz — router o'zi
- eng kichik cost'li shox — OSPF jadvaliga
- cost teng — ikkalasi ham (teng narxli)

11. Area'larga bo'lish

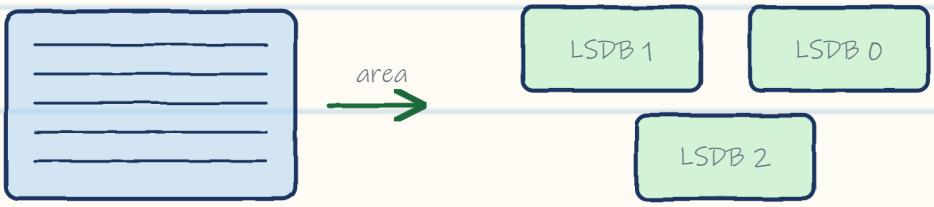


area'lar faqat Area 0 orqali gaplashadi

3-rasm. Area ID — 32 bit; 0 = 0.0.0.0

backbone router	kamida bitta interfeysi Area 0 da
ABR	area'ni Area 0 ga bog'laydi
ASBR	boshqa AS bilan marshrut almashadi
ichki router	hamma interfeysi bitta area da

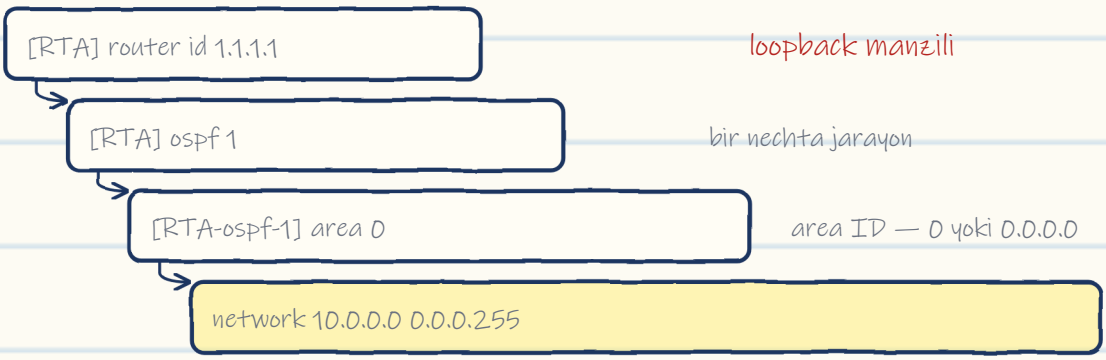
12. Nima foyda



bitta katta LSDB — kichik LSDB lar

Xotira/CPU tejaladi; Hello va LSA area ichida qoladi; ABR da summarizatsiya — LSA kamayadi.

13. Sozlash tartibi



wildcard: 0 — solishtir, 1 — e'tiborsiz

display ospf peer	qo'shnilar, holat, DR/BDR
display ospf lsdb	LSDB — area da bir xil
display ospf routing	cost, next hop, Stub / Transit

Interfeysda: ospf dr-priority · ospf cost. Tiklash: reset ospf process.

Xulosa

- link-state IGP, SPF hisobi — halqa yo'q
- IP protokol 89, multicast 224.0.0.5, VLSM
- Down → Init → 2-Way → Full
- DR/BDR adjacency sonini n(n-1)/2 dan 2(n-2)+1 ga tushiradi
- cost = 100 ÷ polosa (Mbit/s)
- Area 0 — backbone; ABR va ASBR chegarada

Tekshirish

1. OSPF IP protokol raqami? — 89
2. Hello qaysi manzilga? — 224.0.0.5
3. 10 router, DR bilan necha adjacency? — 17
4. 2 Mbit/s interfeys cost? — 50